INNOVASIA 2026 · PROBLEM STATEMENT 12 · DEFECTZERO EDGE

Sandeha

The inspector that knows when it doesn't know.
Electronics inspection from 30 good boards — part by part, in 3D, honest about doubt.



passes on its own — asks a person — rejects on its own

0.995
AUROC on held-out circuit boards it never saw (pcb4, 160 boards)

14.6%
defects escaping vs 39.1% for a typical single-threshold AI, same false-reject rate (12 products)

21 FPS
on 4 laptop CPU threads (ONNX INT8, 224 px). All AI local. Nothing leaves the plant.

Sandeha (संदेह, సందేహ) means *doubt*. Every number in this book was measured by the team on public data (VisA, CC BY 4.0) — none recorded by us, none assumed.

01 · THE PROBLEM

It's 2 a.m. The machine has cried wolf forty times.

An inspector is on her six-hundredth circuit board of the night. Thirty-five of the forty alarms so far were nothing. The forty-first board — the one with the burnt chip — she waves through. It ships.

> This is not a failure of people. It is a failure of machines that never admit when they are unsure.

What PS-12 demands

REQUIREMENT	BAR
Learn a new product	≤ 30 good + ≤ 5 defect images
Calibrated uncertainty	judged on a risk–coverage curve
Human review load	≤ 10% of parts
Domain shift	detect, adapt, no forgetting
Edge speed	≥ 5 FPS
Demo	two lightings, abstain queue, live adaptation

Why today's options fail it

- **Rule-based AOI** must be programmed board by board; AOI vendors themselves report **15–30% false calls** on solder joints (vendor-reported, not independently verified). Operators learn to ignore alarms.
- **Typical AI** outputs one score and one threshold: above it reject, below it pass. It never says *"I don't know"* — so it either scraps good boards or ships bad ones.
- **New product introductions** have no defect data at all: by the time a model has seen enough failures, the boards have shipped.
- **Cloud AI** means sending production images out of the plant — a non-starter for most electronics makers.

What an inspector actually needs is a machine that is right when it is sure, and honest when it isn't — so that when it raises a hand, she believes it.

02 · OUR ANSWER

Doubt is the feature.

Sandeha learns a circuit board from 30 good photos — no labels, no CAD, no training run — finds every component on it, and gives every board one of three answers, with a statistical guarantee on how often a good board is wrongly rejected.

1 **Learn** — 30 good boards → patch memory (DINOv2, frozen) + a golden board

2 **Map** — every component found and typed; its own normal range learned on 40 separate good boards

3 **Ask** — it interviews you about what it is unsure of before it starts

4 **Scan** — unlabelled boards, scored on board pixels only, in ~0.5 s

5 **Decide** — PASS · DOUBT · FAIL

Two calibrated lines, not one threshold

Fail line: a split-conformal bound on good boards — a good board crosses it at most 2% of the time ($\alpha = 0.02$). **Pass line:** set from real and *imagined* defects so few slip under it. Between them, the board goes to a person. A review-budget autopilot keeps that band under 10% — and it may only raise the pass line, never loosen the fail line.

Then it says *why*, part by part

For every rejected board: **which component** (R4, U1, J1…), **what happened** (missing, moved, rotated, burnt, wrong part, solder, damaged, contamination), **where on the line it likely started**, and **how to repair it** — shown on that board's own photo in 3D.

Beyond the brief

- **Approval desk** — a sign-off dossier arguing both sides from the measurements, with cost of either choice and IPC-A-610 grade per fault.
- **Board X-ray** — the board exploded into copper, solder mask, silkscreen and parts, with where failures concentrate.
- **Chip identity check** — reads the marking on every IC to catch wrong, re-marked or burnt chips.
- **Shift guard** — a conformal martingale notices changed lighting or a bumped camera, stops auto-passing, adapts from 5 photos, forgets nothing.
- **Two-person rule** — it only learns when two different reviewers agree; known-answer checks score reviewer reliability.
- **Line context** — line flow, SPC per component, rework tickets, quality report (FPY, DPMO, false calls).
- **All local** — DINOv2, CLIP, Qwen-VL, Depth Anything run on the station.

03 · PRODUCT TOUR — LIVE SCAN

Unlabelled boards in. Answers, with reasons, out.



A real VisA board labelled "missing" by the dataset, scanned live. Verdict FAIL (0.154 > fail line 0.144). The component check names **J1 connector · missing**: only 32% of its footprint is still part body, against 4% variation on good boards. Truth is hidden until the operator asks.

What happens in ~0.5 seconds

STAGE	WHAT IT DOES
Image check	refuses blurred, dark or glare-washed images (→ HOLD, not a guess)
Understand	1,024 DINOv2 patches; board outline aligned onto the photo
Compare	each patch vs its nearest of ~70,000 good-patch memories
Components	30 parts each checked against their own normal range
Locate	regions cut from the heatmap, named, pinned in 3D
Doubt, decide	two calibrated lines → PASS / DOUBT / FAIL

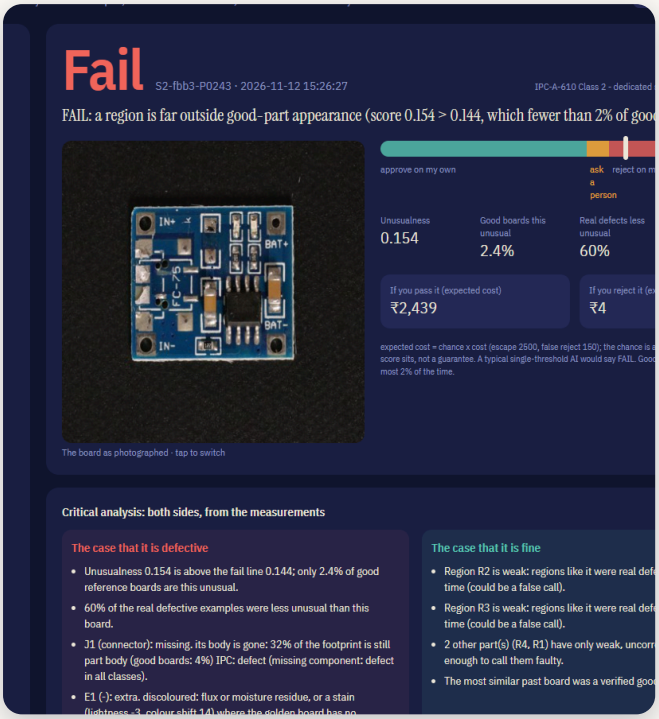
Precision first

A component is called faulty only when two independent measurements agree — its own geometry/colour check *and* the patch-memory heatmap — or the evidence is overwhelming. Everything weaker is a quiet "watch" note.

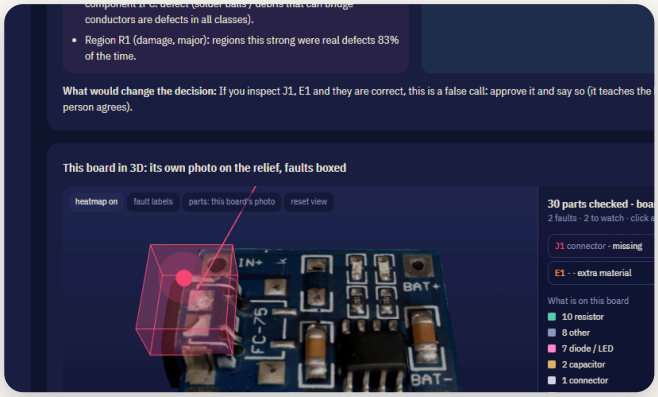
PCB4 · 60 DEFECTIVE + 30 GOOD BOARDS	BEFORE	AFTER
Fault flags on the true defect	44%	**62%**
Wrong fault flags per good board	0.40	**0.07**
Defective boards with a correct flag	97%	93%

04 · PRODUCT TOUR — THE APPROVAL DESK

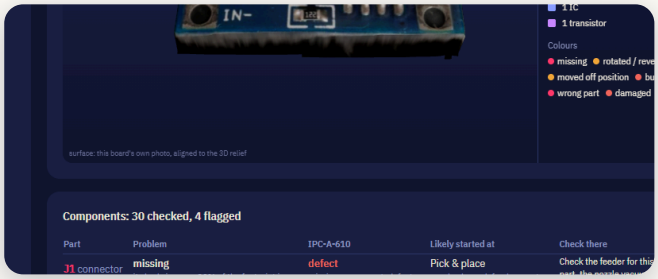
Everything the person at the bench needs to sign.



The dossier: calibrated numbers, the expected cost of passing (₹2,439) vs rejecting (₹4), and a critical analysis that argues **both** sides from measurements — then says what would change the decision.



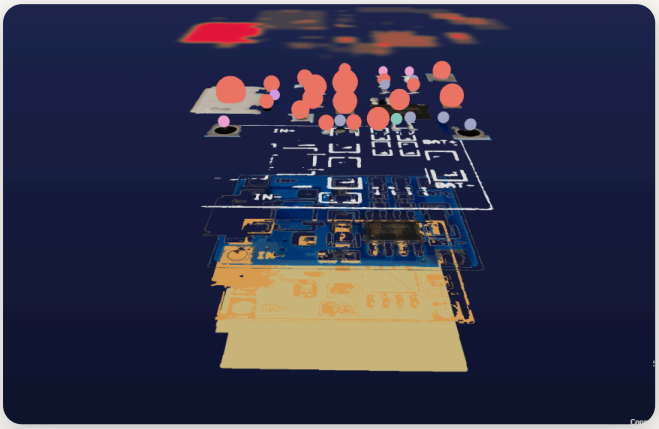
This board in 3D: its own photo draped on a depth relief, the missing connector boxed, every part clickable for type, measurements, cause and repair.



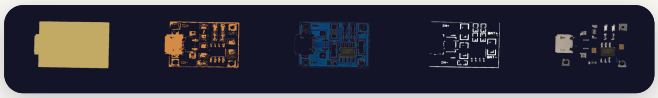
Each fault graded against IPC-A-610 (Class 1/2/3 selectable), traced to the line station where it most likely started, with the repair action.

05 · PRODUCT TOUR — BEYOND THE BOARD

Take the board apart. Read the chips. See the whole line.



Board X-ray. Substrate, copper, solder mask, silkscreen, components and failure density — each layer measured from the golden-board photo, exploded in 3D. Copper read through the mask links parts into a (partial) netlist: click a part to see its impact zone.

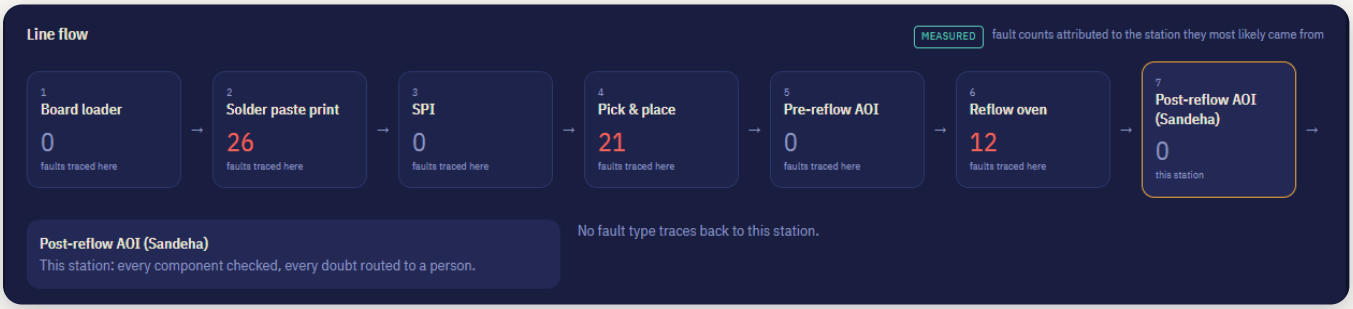


The layers as extracted: board outline, copper, solder mask, silkscreen ink, components.

Chip identity check

BOARD	MARKING READ	VERDICT
Golden (majority of 3 references)	TC4056A	—
Good boards (8 of 8)	TC4056A	match
Burnt boards (5 of 5)	"4056E SN2N1P" / unreadable	different

TC4056A is the real Li-ion charger IC on this board. The local VLM reads text reliably (unlike judging defects), and the comparison is plain string similarity — so the verdict is measurable. Small sample; we say so.



Line flow. Each fault traced to the SMT station where it most likely started (paste print, pick & place, reflow) so the fix goes to the machine, not just the board. Counts here are from all development scans, including before recent fixes.

06 · MEASURED RESULTS

Every number measured on data we never trained on.

0.934
mean AUROC, 12 VisA products × 3 seeds (672 px)

14.6%
escapes vs **39.1%** single threshold, same 3.9% false rejects

10.2%
review rate at 3% defect prevalence

21 FPS
ONNX INT8, 224 px, 4 CPU threads (8.5 FPS at 336 px)

Board-only scoring on the charger board (pcb4, 80 good + 80 defective held-out boards)

MEASURE	WHOLE PHOTO	BOARD ONLY (SHIPPED)
AUROC	0.986	**0.995**
Good boards wrongly rejected	2.5%	**0%**
Defective boards passed	0 of 80	**1 of 80**
Boards sent to a person	20.6%	**10.6%**
Callouts landing off the board	23.7%	**2.5%**

Callouts and components

MEASURE (VISA)	RESULT
Region callouts on the real defect (4 PCB-type products)	75%
Defective parts with a callout on the true defect	90%
Components mapped on pcb4, no labels	30
Component-type classifier, held-out photos	86.7%
Real "missing" parts diagnosed (pcb4)	87%
Defective boards localised, all 4 PCBs	**78%**

Robustness

MEASURE	RESULT
Shift (camera bumped / light change) detected	within 10–15 parts
Adaptation from 5 photos	AUROC 0.70 → 0.92
Forgetting after adaptation	none (probe set unchanged)
Scan latency on the station	~0.4–0.7 s per board
Review-budget autopilot (pcb4)	review ~40% → 11%

The type classifier was first scored at 95%; we found 12 of its 120 held-out photos duplicated in training, removed them, and report the honest 86.7%.

07 · HOW IT COMPARES

What others give you, and what Sandeha gives you.

CAPABILITY	RULE-BASED AOI	TYPICAL ANOMALY AI	CLOUD VISION MODEL	SANDEHA
Data to start	CAD + programming per board	hundreds of images	none, but no calibration	**30 good photos, no labels**
Says "I don't know"	no	no (one threshold)	no	**yes — calibrated doubt band**
Guarantee on false rejects	no	no	no	**yes — conformal, α = 2%**
Explains which part, why	per programmed rule	a heatmap	free text, unverified	**component · fault · cause · repair · IPC grade**
Notices changed conditions	no	no	no	**conformal martingale alarm**
Adapts without forgetting	re-program	retrain	n/a	**5 photos, memory only grows**
Learns from people safely	no	often blindly	no	**two-person rule + reviewer checks**
Wrong / burnt chip	if programmed	sometimes	sometimes	**marking read and compared**
Data leaves the plant	no	depends	**yes**	**never — all models local**
Runs on edge	dedicated machine	usually GPU	no	**21 FPS on laptop CPU**

The number that matters

At the **same** false-reject rate, a single-threshold model let **39.1%** of defects escape across 12 products. Sandeha let **14.6%** escape — because instead of guessing on hard cases, it asks a person about ~10% of parts.

Why judges can trust the numbers

Public data (VisA, CC BY 4.0), held-out test sets, three seeds, leakage removed when found, and every limitation printed on screen next to the feature it limits. Nothing in this book is a projection.

08 · WHAT IT CAN'T DO YET

We tell you the limits, because that is the whole idea.

Measured limitations

- **Dry solder joints** are not detectable from a single top-down photo: normal solder shine varies more than a dull joint changes it. Real AOI uses angled, multi-colour ring lights; so would we.
- **Denser boards are harder.** The component check is strongest on the charger board (pcb4); across all four PCB products, 78% of defective boards are localised.
- **One escape in eighty** on pcb4 with board-only scoring — the price of halving the review load and removing false rejects.
- **Defect names from 5 examples** are barely better than guessing (52% vs 49%): shown as a guess.
- **The VLM is a second pair of eyes.** The 2B model was no better than chance at confirming defects, so it never changes a verdict; it reads chip markings and describes close-ups.
- **Connectivity is partial** — vias and inner layers are invisible from one photo.
- **Simulated, labelled:** feeder/reel data, moisture-sensitive part clocks and the reflow profile (the camera cannot see them).

Next, in order

- Angled RGB ring light → dry joints, tombstoning, lifted leads.
- Tune component segmentation on dense boards (headers, transducers, through-hole).
- Qwen2.5-VL-7B (4-bit, on the station) for richer descriptions; re-measure before trusting it.
- Two cameras or a tilt stage → true 3D instead of a monocular relief.
- Connect real SPI, pick-and-place and oven data (IPC-CFX / Hermes) to replace the simulated line context.
- Raspberry Pi 5 measurement (not yet measured).

> A system that hides its weak spots will eventually surprise you on the line. One that shows them lets you plan around them.

09 · PITCH SCRIPT (≈ 4½ MINUTES)

Spoken by two: one for the story, one for the demo.

0:00 — HOOK · SLOW · DARK SLIDE

It's 2 a.m. on an electronics line. An inspector is looking at her six-hundredth circuit board of the night. The inspection machine beside her has raised an alarm forty times this shift. Thirty-five of them were nothing.

So when the forty-first alarm goes off, she does what any tired human would do. She waves it through. That board had a burnt chip. It shipped. *[pause]*

This isn't a failure of people. It's a failure of machines that never admit when they are unsure.

0:35 — THE PROBLEM · SLIDE "DEFECTZERO EDGE"

Our problem statement asks for something hard and honest: learn a new product from thirty good photos and five defects; know when you are unsure; send no more than ten percent of parts to a person; survive a change in lighting; run on the edge.

Today's options fail this in two ways. Classic inspection machines must be programmed board by board — and their own vendors report that fifteen to thirty percent of their alarms are false. Typical AI models give you one number and one threshold. Above it, reject. Below it, pass. They never say "I don't know." So they either scrap good boards, or ship bad ones.

1:15 — WHAT WE BUILT · LOGO SLIDE → DEMO

We built Sandeha. In Sanskrit, sandeha means doubt. **Doubt is the feature.**

We show it thirty good boards. No labels, no CAD files, no training run. In about a minute it finds every component on the board — the USB connector, the IC, the resistors, the capacitors — and learns how each one normally looks.

Now boards arrive unlabelled. Not even this screen knows which are bad. Every board gets one of three answers: pass on its own, reject on its own — or the one nobody else gives you: "I'm unsure — a person decides."

And when it rejects, it doesn't just say "defect". It says which part, what happened, where on the line it likely started, and how to repair it — on this board's own photo, in 3D.

2:15 — WHY IT'S BETTER · NUMBERS SLIDE · LET THEM LAND

Across twelve products in a public industrial dataset, at the same false-reject rate as a single threshold, a typical AI let **thirty-nine percent** of defects escape. Sandeha: **fourteen point six** — while asking a person about only ten percent of parts.

On the charger board you just saw, scored on a hundred and sixty boards it had never seen: zero good boards wrongly rejected. One defect in eighty got through. About one board in ten sent to a person.

[Approval desk] When it asks, it doesn't just hand over a problem. The inspector gets a full dossier: the case that the board is defective, the case that it's fine, the cost of each choice, and the industry acceptance grade for every fault.

[Board X-ray] It pulls the board apart into its real layers — copper, solder mask, print, parts — and shows where failures have concentrated over the whole shift. And it reads the part number printed on each chip: on every good board it read TC4056A; every burnt board we checked read differently.

If the camera is bumped or the lights change, it notices within ten to fifteen boards, stops approving on its own, adapts from five good photos, and forgets nothing it knew. It runs at twenty-one frames per second on a laptop CPU. Every AI model runs on the station. No cloud. Nothing leaves the plant.

3:40 — HONESTY · SLIDE "WHAT IT CAN'T DO YET" · CALM

We'll tell you what it can't do, because that's the whole idea. It cannot see a dry solder joint from one camera above the board — real machines use angled coloured lights for that, and so would we. Its part-level checks are strongest on the board you saw, and weaker on denser boards. Its AI advisor is a second pair of eyes, never the judge. We measured all of this ourselves, and we show it on screen.

4:05 — CLOSE · DARK SLIDE · SLOW

Remember the inspector at 2 a.m.? She doesn't need a machine that's always confident. She needs one that's **right when it's sure**, and **honest when it isn't** — so that when it raises a hand, she believes it.

That's Sandeha. The inspector that knows when it doesn't know. Thank you.

Delivery notes

- Two-second pauses after "It shipped.", "Doubt is the feature." and the escape numbers.
- Hand-over from story-teller to demo-driver at "We built Sandeha."
- Short on time? Cut the X-ray paragraph first, then the drift line. Never cut the honesty section.
- Quote numbers only as worded here. Attribute "15–30% false calls" to AOI vendors' own reports.

Run-sheet (≈ 3 minutes inside the pitch)

TIME	SCREEN	DO / SAY
1:20	Choose a board	pcb4 (Li-ion charger) — already learned
1:30	Scan boards	"Scan the whole tray": unlabelled boards, stages with real timings
1:50	A FAIL	point at the component check and the 3D box on its own photo
2:30	Approve	open the dossier: case for / against, cost, IPC grade, root cause
3:00	Board X-ray	"Explode", click the most-failed part, read a chip
3:20	Demo mode	scenario 4 (camera bumped) if time: the alarm and adaptation

Before going on stage

- Start the station; wait for "Mapped N components" and "Recalibrated on the board only".
- Scan one tray in advance so the GPU and models are warm.
- Keep one known FAIL board's link ready: `?session=…&part=…#approval`.
- Have the honesty slide on screen during Q&A.

Likely questions

Isn't 30 photos too few?
For a nearest-neighbour patch memory, no: it compares every patch to what good looks like. The guarantee is on the good side (≤2% false rejects); escapes are measured, not promised.

What if the lighting changes?
A conformal martingale detects it within 10–15 boards; auto-pass is suspended; 5 photos adapt it; a fixed probe set proves nothing was forgotten.

Why trust the AI explanations?
The verdict comes from calibrated measurements. Explanations are composed from those measurements and cited; the VLM only describes and reads text.

How is this different from AOI?
No board programming, a calibrated "I'm unsure", per-component normal ranges learned from data, and every limit shown.

Is the data real?
Yes — VisA (Amazon, CC BY 4.0): real boards, real defects with pixel masks. Component photos MIT; 3D models KiCad; lighting CC0.

What does it cost to run?
A laptop-class CPU reaches 21 FPS at 224 px; the GPU is only needed for the larger optional models.

11 · DATA, MODELS, LICENCES, SOURCES

Data

- **VisA** (Amazon, AWS Open Data) — 12 products incl. 4 PCB types, real defects with pixel masks — CC BY 4.0.
- **MAD-Sim** (PAD, NeurIPS 2023) — posed multi-view objects for the 3D twin — CC BY-NC-SA 4.0.
- **PCB Component Classifier** (ParsaVictor, Hugging Face) — 1,141 photographed SMD parts after de-duplication — MIT.

Models (all run on the station)

- DINOv2-S/14 (Meta) — frozen patch features, patch memory.
- CLIP ViT-B/16 (OpenAI) — region naming fallback.
- Qwen2-VL-2B / Qwen2.5-VL-7B (Alibaba, Apache-2.0) — descriptions, chip markings.
- Depth Anything V2 Small — single-photo 3D relief.
- 3D component models: KiCad Packages3D — CC BY-SA 4.0 with library exception. Lighting: Poly Haven, CC0.

Methods

- Split-conformal two-threshold policy; review-budget autopilot.
- Conformal test martingale for shift; probe-set forgetting check.
- Per-component robust limits (median ± 5 robust SD) on 40 separate good boards.

Standards and industry sources

- IPC-A-610 side overhang limits (secondary sources): pcblibraries.com/Forum/topic3111.html · smtnet.com/library/files/upload/IPC%20training.pdf · atlaspcb.com/blog/ipc-a-610-acceptance-criteria-inspection-classes-guide
- IPC-7912 DPMO structure: ipc.org/TOC/IPC-7912A.pdf · pcbsync.com/ipc-7912/ · factorylogix.atlassian.net/wiki/x/4qILC
- J-STD-033 floor life: anglia-live.com/pdfs/MoistureSensitivityReference.pdf · smtnet.com/library/files/upload/01-ipc-jedec-j-std-033-msd-storage.pdf
- SMT defect root causes: protoexpress.com/blog/common-errors-surface-mount-technology-smt/ · pcbgogo.com (SMT defects guides)
- AOI false-call rates (vendor-reported): iconnect007.com/article/51662 · siemens.com (Opcenter AOI FCR) · ifactoryapp.com (AI vision vs AOI)

Reproduce our numbers

`scripts/p1_eval.py · region_eval.py · board_eval.py · flag_precision.py · board_only_eval.py · train_components.py · export_onnx.py`

Results land in data/results/. Tests: 25 backend tests (pytest).